Your fundamental questions are?

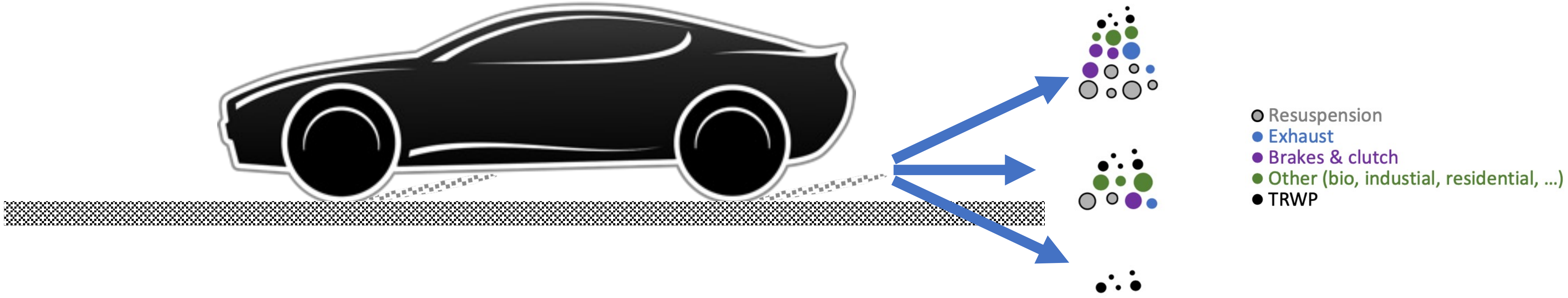
- Design the optimum passenger car tyre tread compounds that improves tyre wear properties.
  - Focus on which polymer or polymer blend and which fillers or other additives are required.
- Describe how you would mix the rubber compound and then prepare it for moulding into a car tyre.
- Explain how your tyre would perform in the other tyre performance measures such as enhanced fuel economy or emergency braking tests.

Case study 2nd April 2025
Room Eng.209

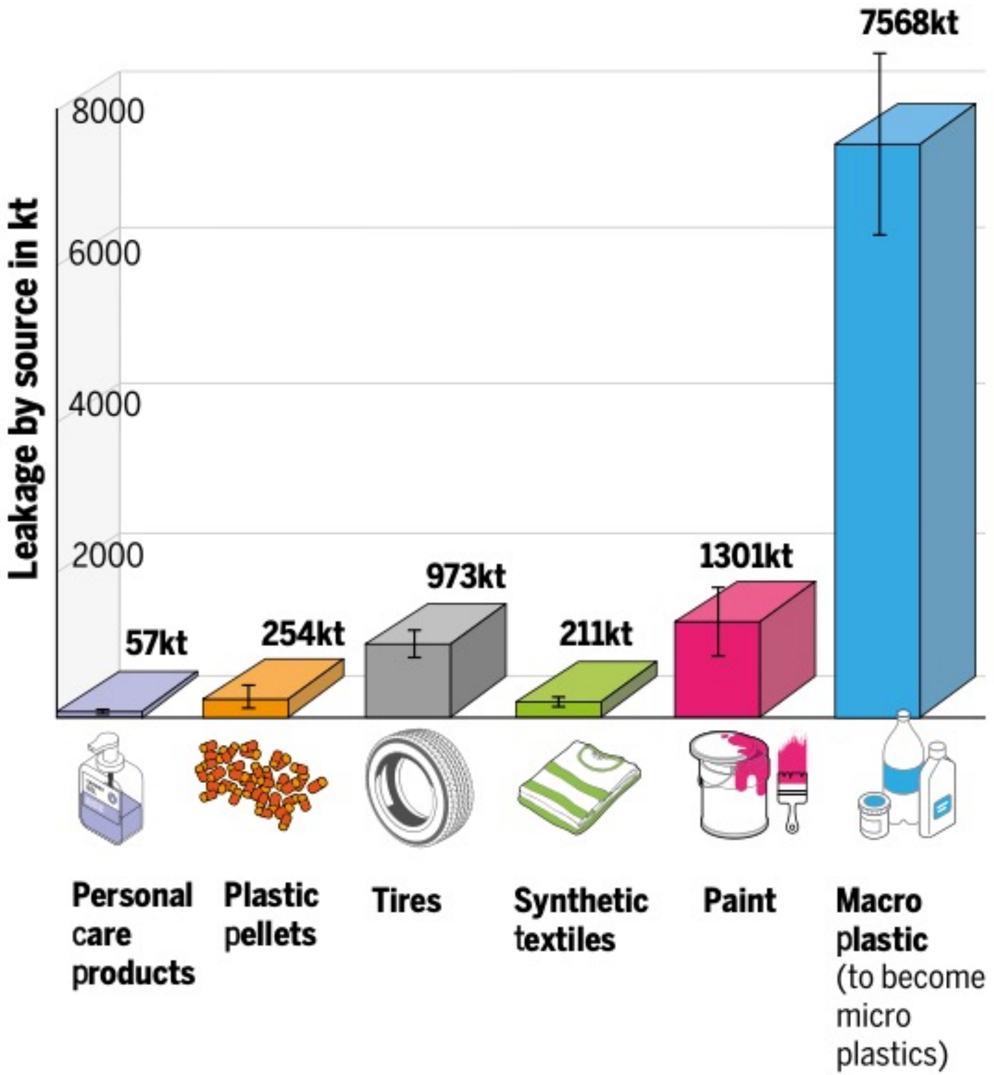
Tyre Road Wear Particles TRWP

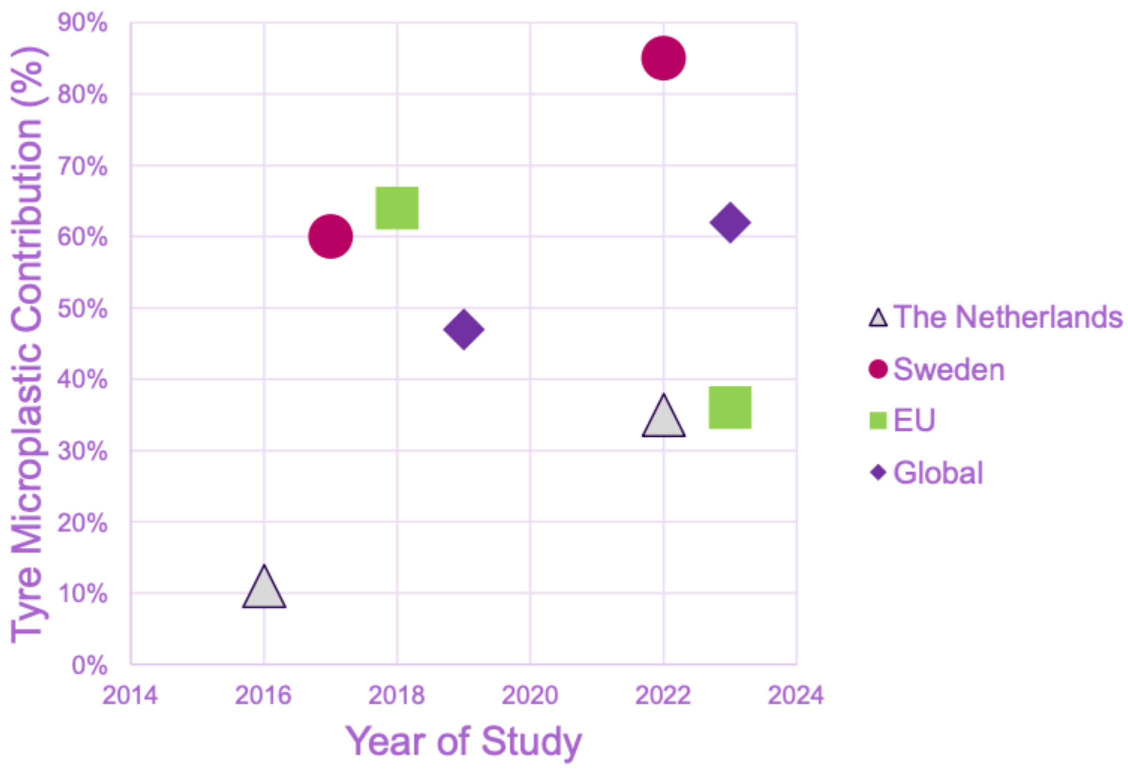
Prof. James Busfiled

Prof. Keizo Akutagawa



Micro Plastics and Tyre Road Wear Particles (TRWP)



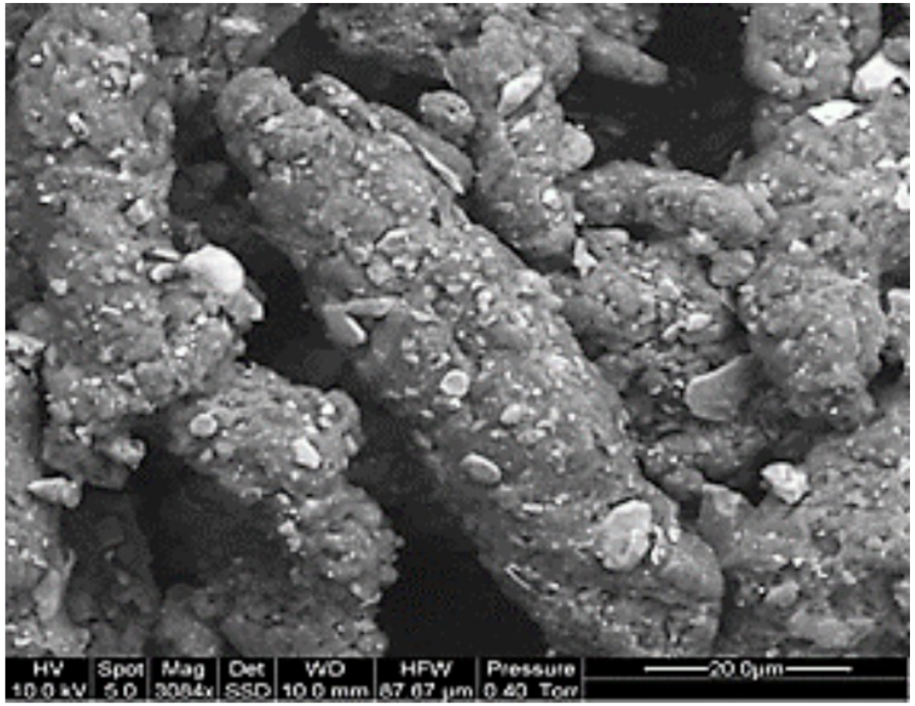


Source: Richard C. Thompson et al. Science 386, eadl2746 (2024).

Giechaskiel, B., et al. Sustainability 2024, Vol. 16, Page 522, 16(2), 522

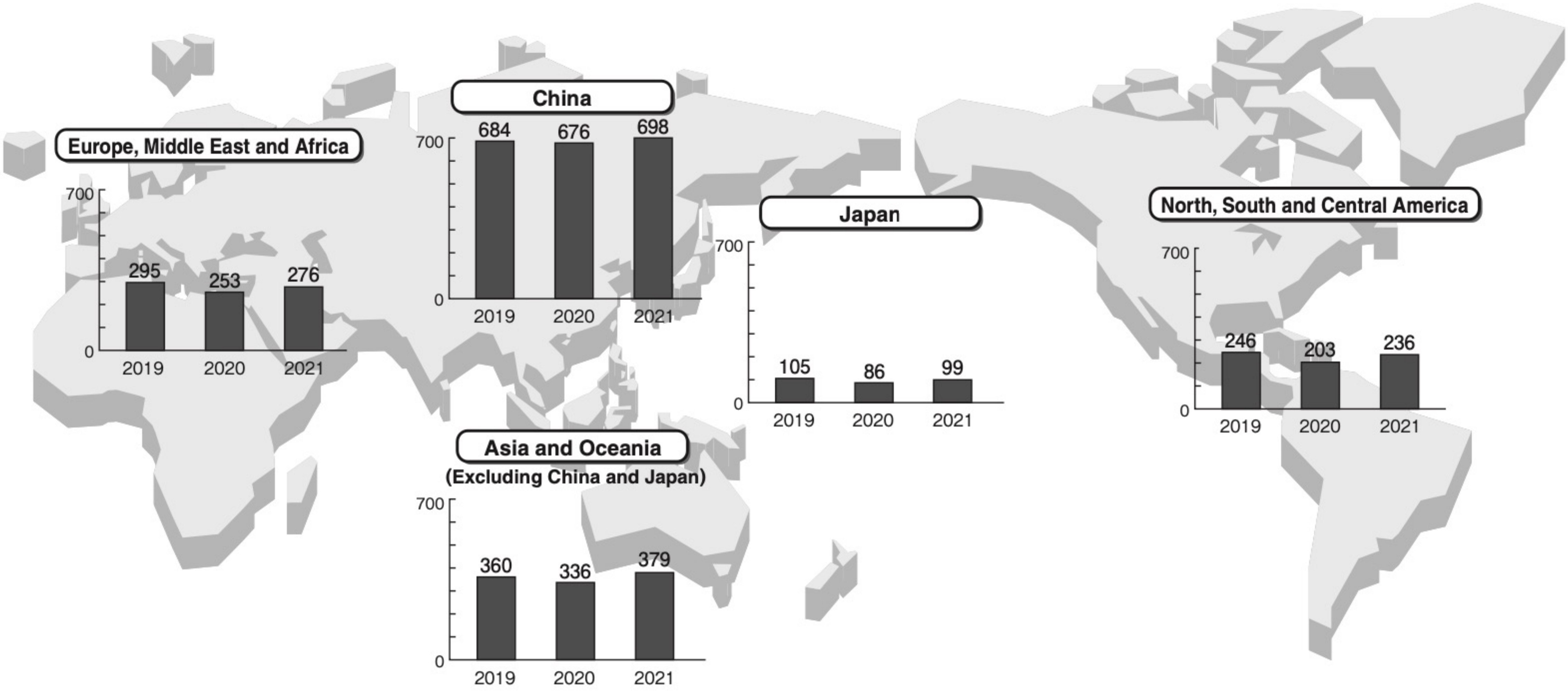
What is Tyre Road Wear Particles (TRWP) ?

1. Emitting due to the friction occurring between the vehicle tyres and the road surface
2. Mixture of tyre tread material and pavement material in equal parts
3. Cigar shaped particles, generally below 1mm in length, typically between 70-100um
4. Average density of 1.8



Source: :Summary note, Workshop on Microplastics from Tyre Wear: Knowledge, Mitigation Measures, and Policy Options, Monday 18 – Wednesday 20 May 2020

Tyre Production Worldwide



Total = 17 million tons of tires / year

N.B.: 1. Unit: x10,000 tons (produced rubber)
2. Including tyres other than vehicle tyres.

Source: IRSG "The World Rubber Industry Outlook" (December 2021)

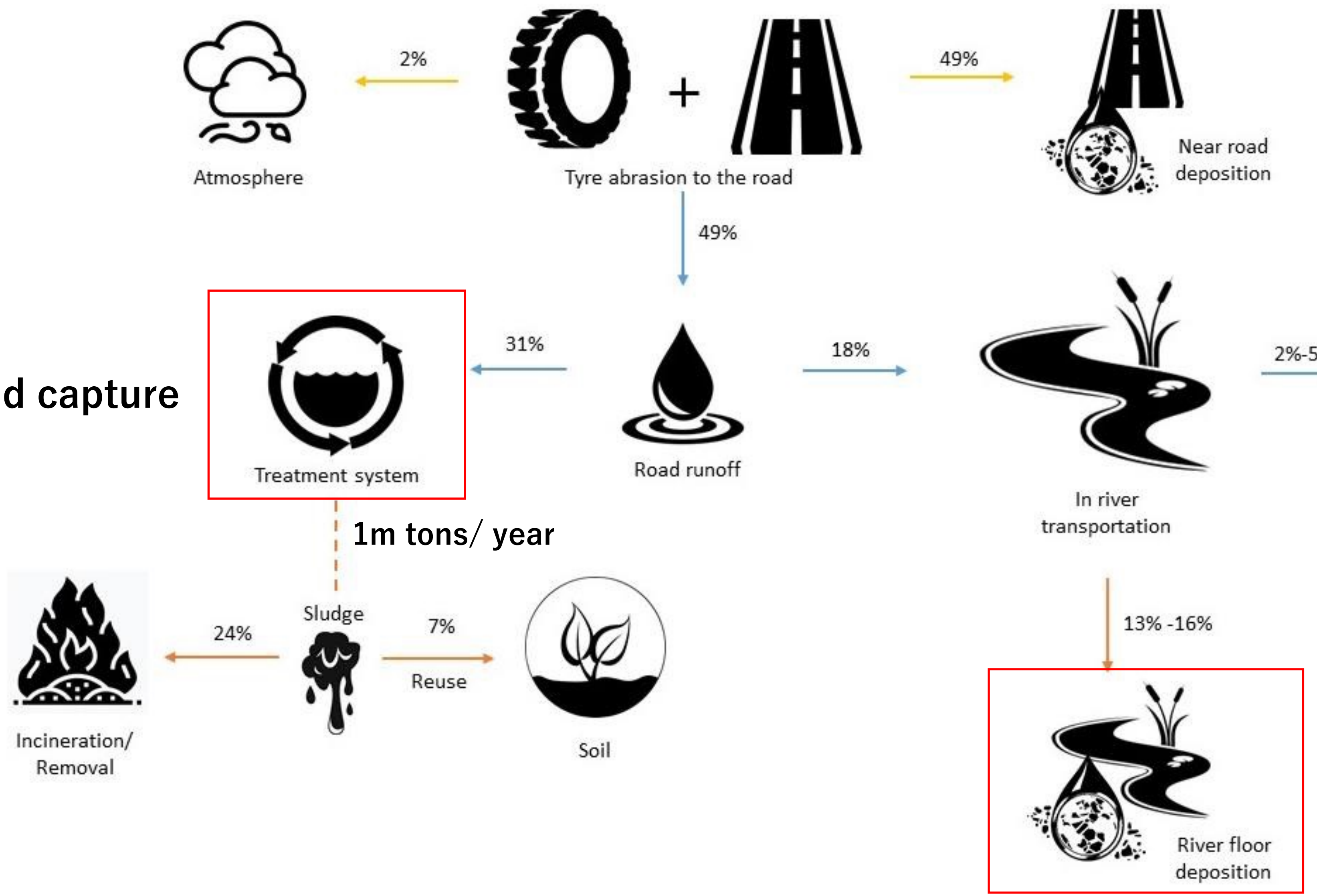
Total Emission of Wear Debris from Tyres (tons/year)

	Number of Capita [42]	Number of Cars [52]	Total Emission from Tyres (tonnes/year)	Emission per Capita/year (kg)
The Netherlands	17,016,967	9,612,273	8834	0.52
Norway	5,265,158	3,671,885	7884	1.5
Sweden	9,880,604	5,755,952	13,238	1.3
Denmark	5,593,785	2,911,147	6721	1.2
Germany	80,722,792	52,391,000	92,594	1.1
United Kingdom	64,430,428	35,582,650	63,000	0.98
Italy	62,007,540	51,269,218	50,000	0.81
Japan	126,702,133	76,763,402	239,762	1.9
China	1,373,541,278	250,138,212	756,240	0.55
India	1,266,883,598	159,490,578	292,674	0.23
Australia	22,992,654	17,180,596	20,000	0.87
USA	323,995,528	265,043,362	1,524,740	4.7
Brazil	205,823,665	81,600,729	294,011	1.4
Total	3,564,856,130	1,011,411,004	3,369,698	0.95

Total 3.4 million tons/ year = 20% x 17 million tons of tires / year

Source: Pieter Jan Kole, et al, International Journal of Environmental Research and Public Health **17**, 1265 (2017)

Fate of Tyre Road Wear Particles (TRWP)



③Filter and capture

1m tons/ year

100k tons/ year

500k tons/ year

① Remove Toxicity

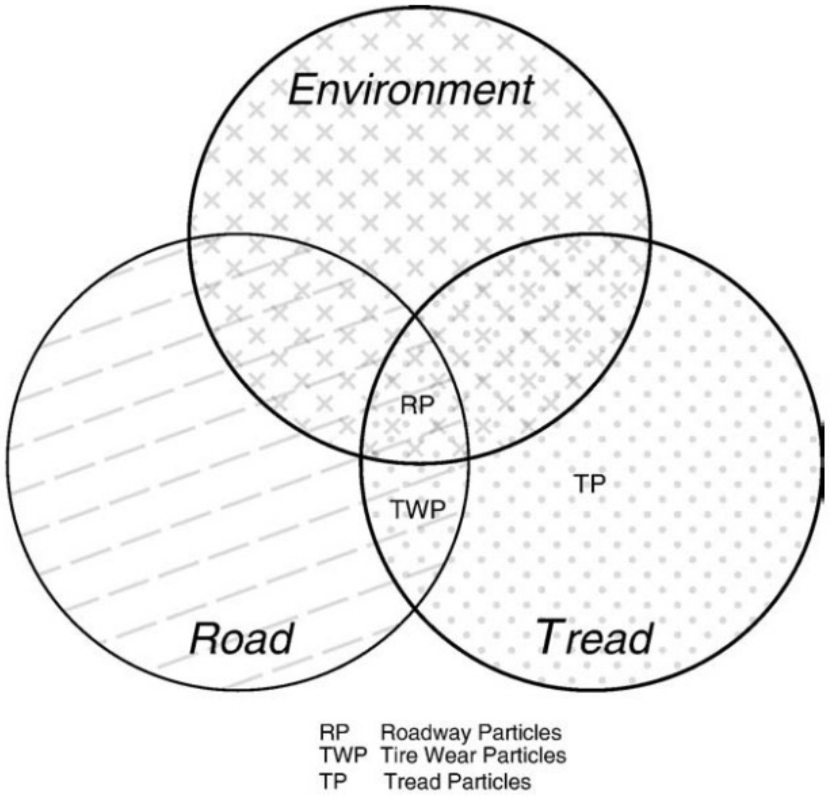
② Reduce TRWP

Source: EUROPEAN TRWP PLATFORM, Way Forward Report (2019)

Composition of Tyre Road Wear Particles (TRWP)

Chemical family	RP	TWP	TP
Plasticizers and oils	13	10	19
Polymers	23	16	46
Carbon blacks	11	13	19
Minerals	53	61	16

Values are expressed in percent by mass.



Polycyclic Aromatic Hydrocarbons (PAH)

Chemical	RP	TWP	TP
Acenaphthene	4.08	0.04	0.13
Naphthalene	6.1	0.2	1.18
Phenantherene	53.4	1.66	1.21
Pyrene	54.84	4.77	0.06
Acenaphthalene	0.14	0.15	1.24
Anthracene	7.36	0.1	0.11
Benzo(a)Anthracene	38.65	0.18	2.87
Benzo(a)pyrene	12.51	0.28	N.D.
Benzo(b)fluoranthene	7.4	0.37	0.92
Benzo(g,h,i)perylene	4.04	3.22	1.77
Benzo(k)fluoranthene	7.4	0.02	0.92
Chrysene	17.72	0.36	2.95
Dibenzo(a,h)anthracene	2.56	0.1	0.87
Fluoranthene	82.13	0.98	1.62
Fluorene	1.76	0.07	0.25
Indeno-1,2,3(c,d)pyrene	5.36	0.21	N.D.
Total	305.45	12.71	16.1

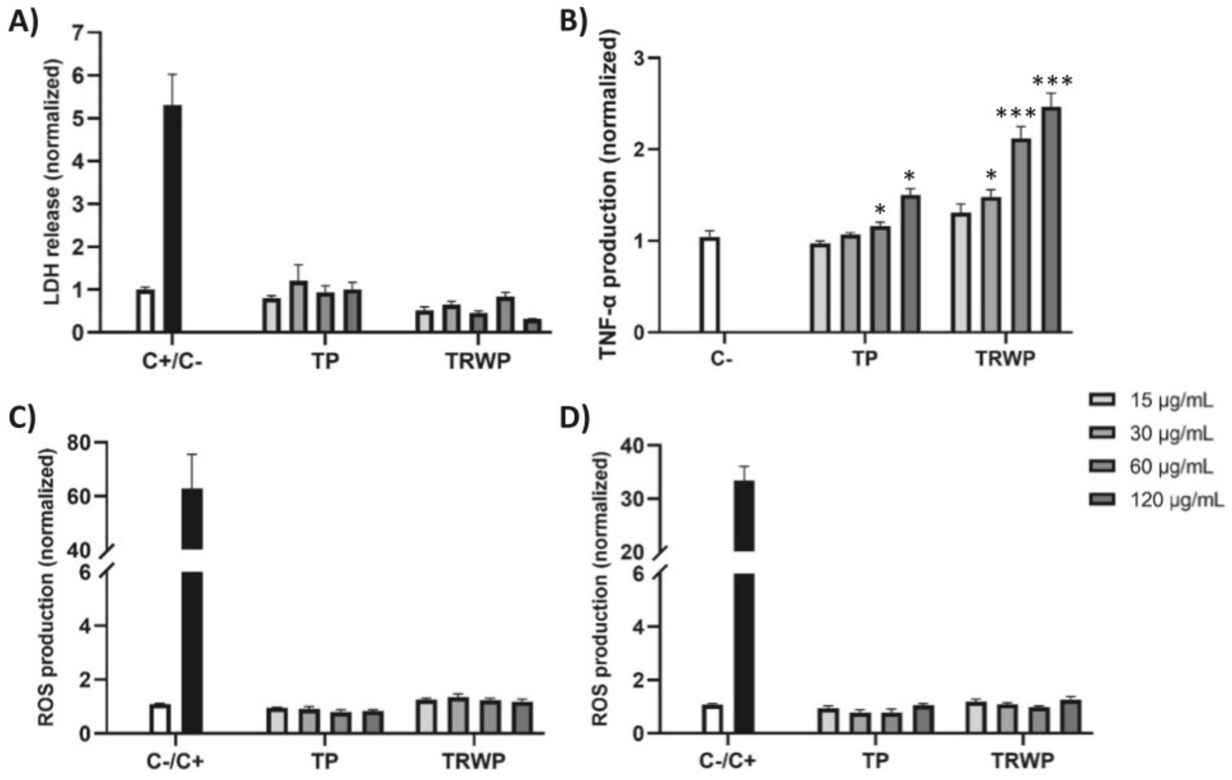
Metal	RP	TWP	TP
Used in tire manufacturing			
Zinc	4000	3000	9000
Silicon	86,000	87,000	54,000
Sulfur	9000	5000	12,000

Values are expressed in parts per million (ppm) of particle mixture.

Source: Marisa L. Kreider et al. Science of the Total Environment 408 (2010) 652–659

Toxicological profile of of Tyre Wear Particles Leachate

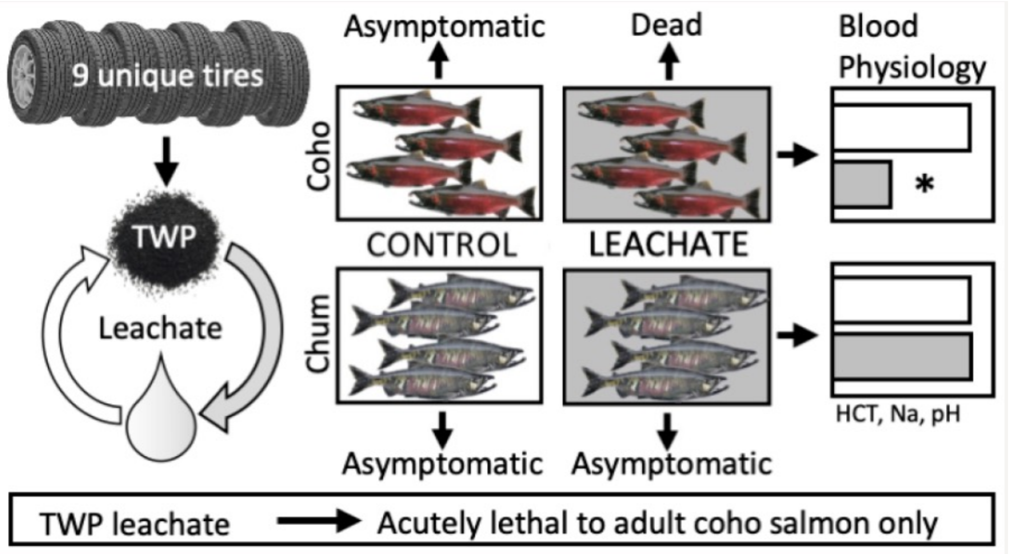
- Human health



Both TP and TRWP induced neither significant cytotoxicity nor oxidative stress, but triggered a concentration-dependent proinflammatory response, as evidenced by increased TNF-α production

Source: Abderrahmane Bouredji, et al. Environment International 190 (2024) 108885

- Fish mortality

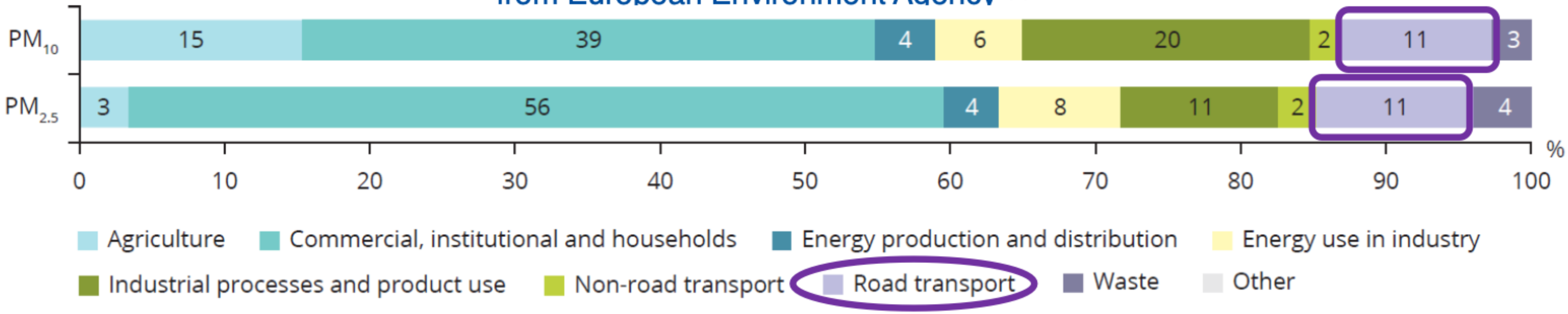


Future research will invariably address the occurrence of 6PPD-quinone and other TWP-derived chemicals in aquatic environments and begin assessing additional species for adverse outcomes.

Source: McIntyre, J. K., et al. Environmental Science and Technology, 55(17), 11767–11774 (2021)

Size of Tyre Road Wear Particles (TRWP) and Environment

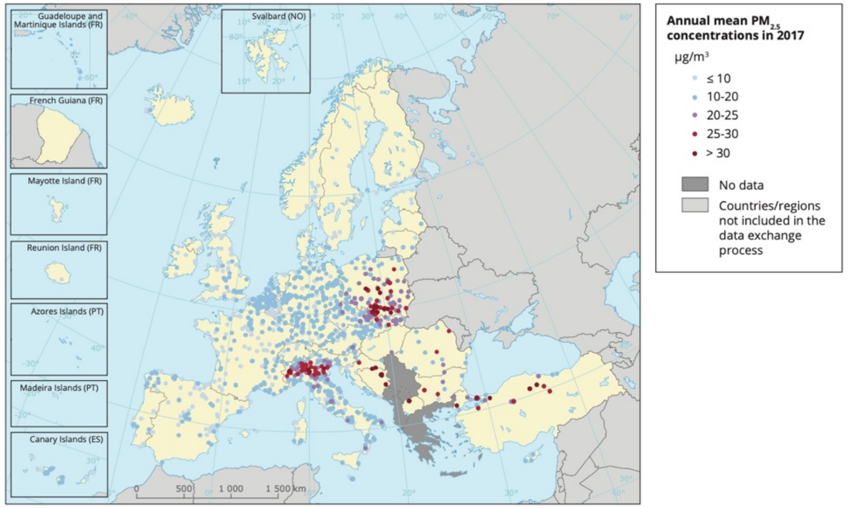
Contribution to EU-28 PM emissions from the main source sectors in 2017, from European Environment Agency [1]



Pollutant	Averaging period	Standard type and concentration	Comments
PM_{10}	1 day	EU limit value: 50 µg/m³	Not to be exceeded on more than 35 days per year
		WHO AQG: 50 µg/m³	99th percentile (3 days per year)
	Calendar year	Limit value: 40 µg/m³	
		WHO AQG: 20 µg/m³	
$PM_{2.5}$	1 day	WHO AQG: 25 µg/m³	99th percentile (3 days per year)
	Calendar year	EU limit value: 25 µg/m³	
		EU exposure concentration obligation: 20 µg/m³	Average exposure indicator (AEI) ([a]) in 2015 (2013-2015 average)
		EU national exposure reduction target: 0-20 % reduction in exposure	AEI ([a]) in 2020, the percentage reduction depends on the initial AEI
		WHO AQG: 10 µg/m³	

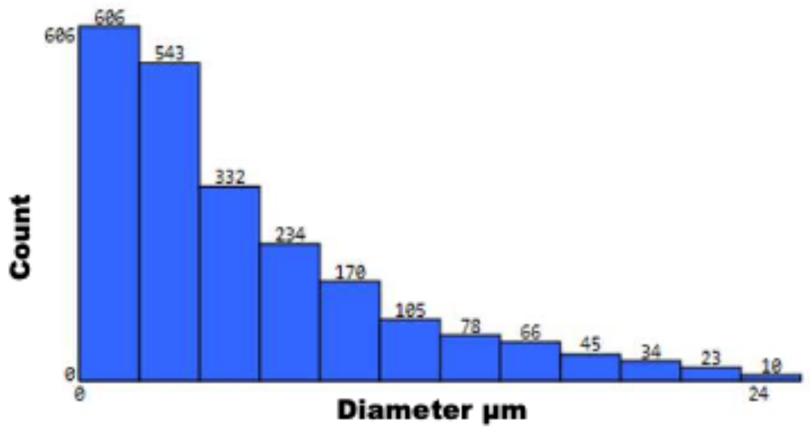
Upper respiratory tract (nose, throat, and bronchi) and respiratory illnesses

Deep into the lungs and bloodstream, increasing the risk of respiratory and cardiovascular diseases



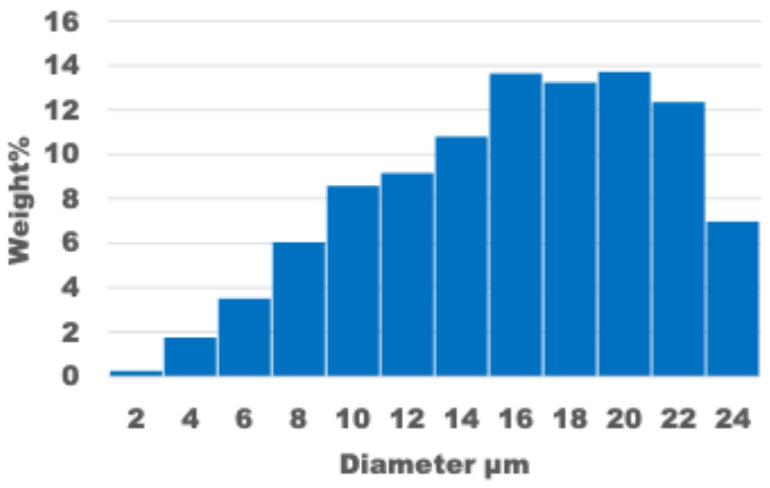
Source: EEA. (2019). Air quality in Europe — 2019 report. https://www.eea.europa.eu/publications/air-quality-in-europe-2019

Size and Volume effect of Tyre Road Wear Particles (TRWP)



Small particles have the largest count.
This histogram is important for inhalation study.

PM 2.5 emission

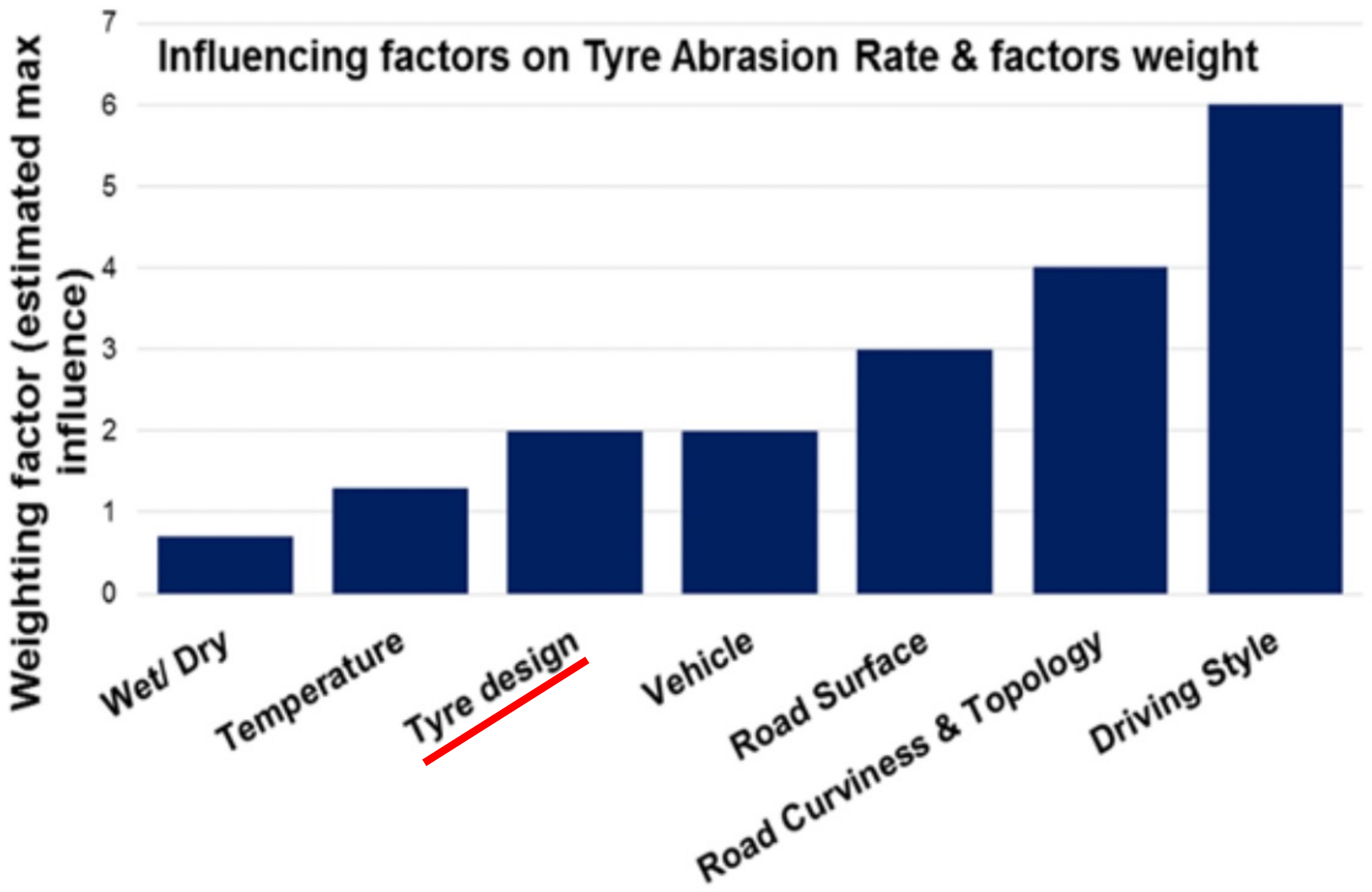


Weight % of the small particles is almost negligible.
This histogram is important for leaching study.

Toxicity of leaching

Source: Hiroshi Mouri, Tire Technology Expo 2025 d2_s4_p8

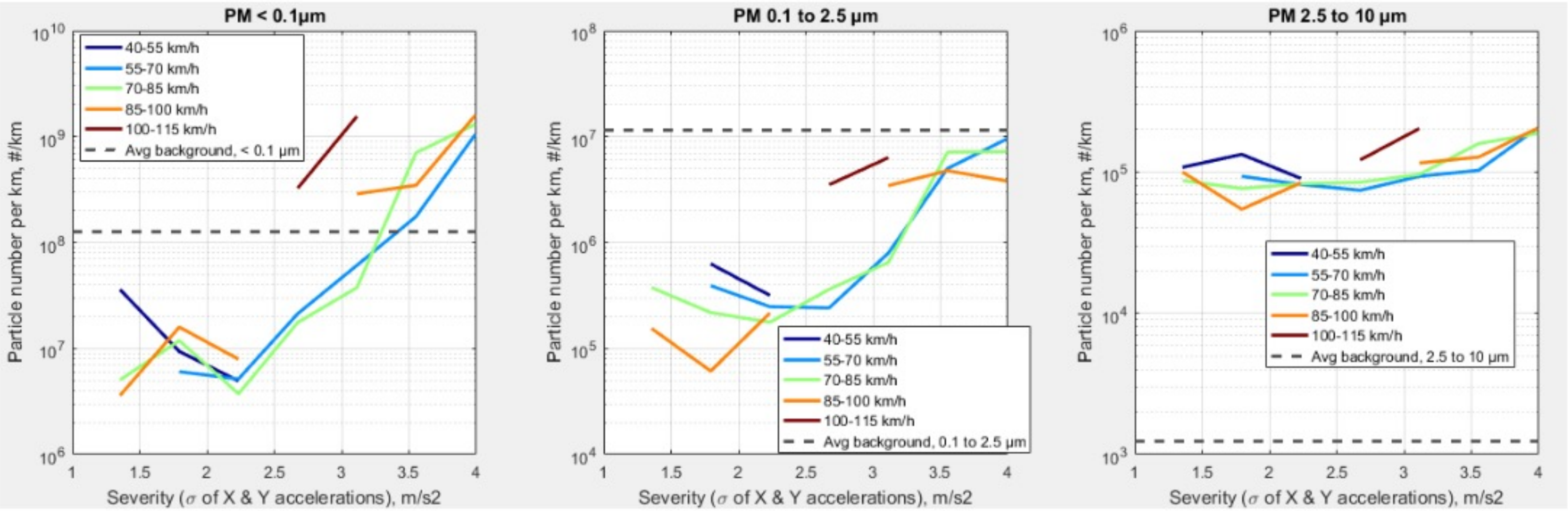
Factors Affecting Tyre Wear Rate



- **Tyre characteristics:** Size, tread depth, construction, tyre pressure and temperature, contact patch area, chemical composition, accumulated mileage.
- **Road surface characteristics:** Pavement construction, aggregate rocks, binder (bitumen, cement), porous asphalt, macro and micro texture, porosity, condition, road surface wetness, road dust loading in surface texture.
- **Driving behavior/Vehicle operation:** Speed, linear and radial acceleration, frequency and extent of braking and cornering.
- **Vehicle characteristics:** vehicle weight and distribution of loads, location of driving wheels, wheel alignment, engine power, power/unassisted steering, electronic braking systems, suspension type and condition.

Effect of Driving behavior/Vehicle operation on Size of TRWP

Same conclusion if we look at the different sizes : severity is much more important than speed



- Tire wear is proportional to Friction Energy :

$$Wear \cong Friction\ Energy = Friction\ Force \times Sliding\ distance$$

$$Friction\ Force = Mass_{Vehicle} \times Acceleration\ of\ vehicle_{longitudinal\ \&\ lateral}$$

Depends on Tire Design & Friction Force

Source: Frederic Biesse, Tire Technology Expo 2023 d1_s3_p1

Type of Worn Surface of Tyres: Searing and Abrasion

Conventional vehicle

Low severity

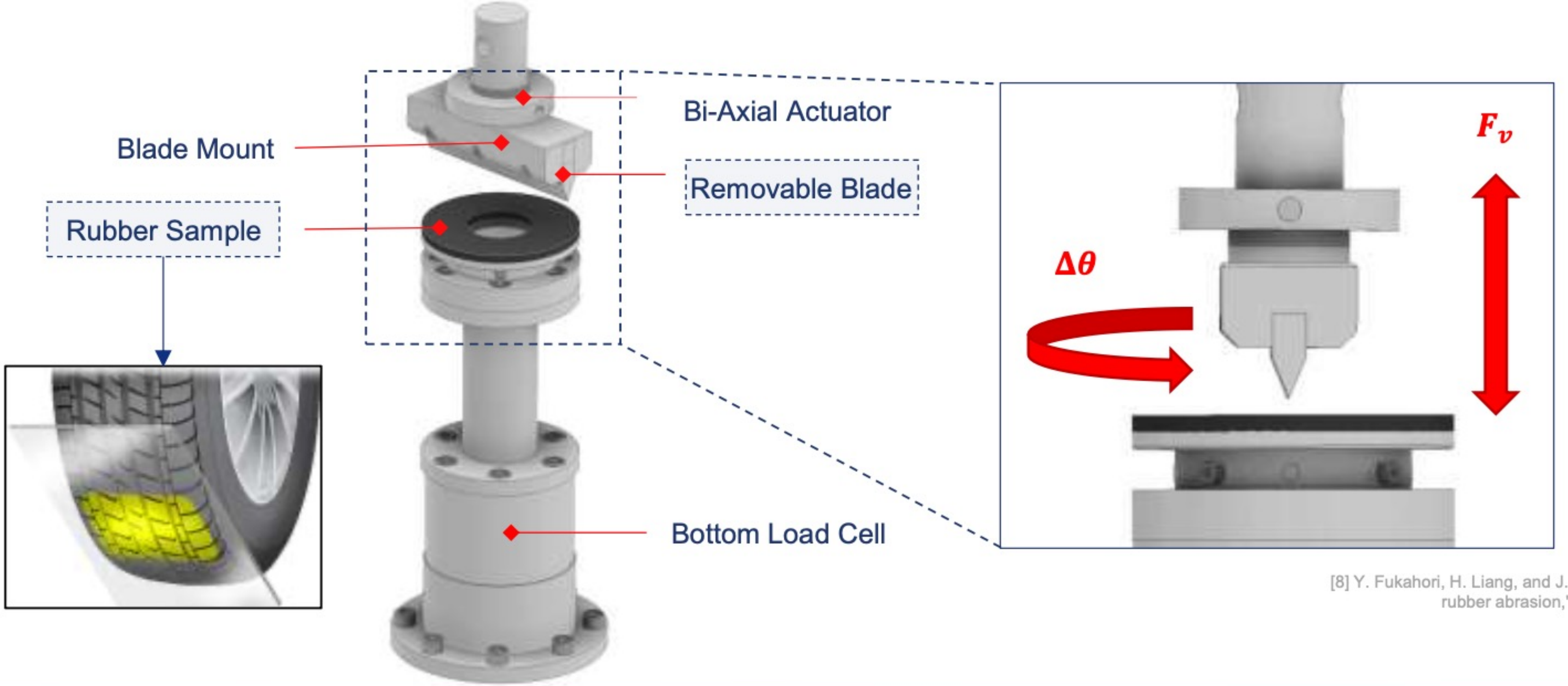
No abrasion pattern (smearing)

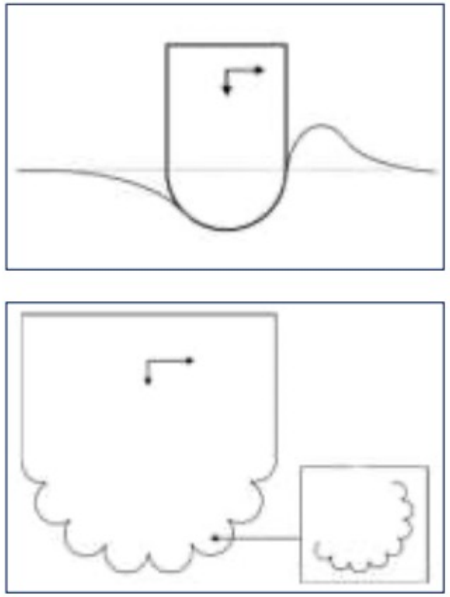
High performance vehicle

High severity

Small abrasion pattern

Carry out analysis of viscoelastic properties of tread compounds → Prepare samples for abrasion configuration → Subject samples to uni-directional dynamic abrasion (UDM)*





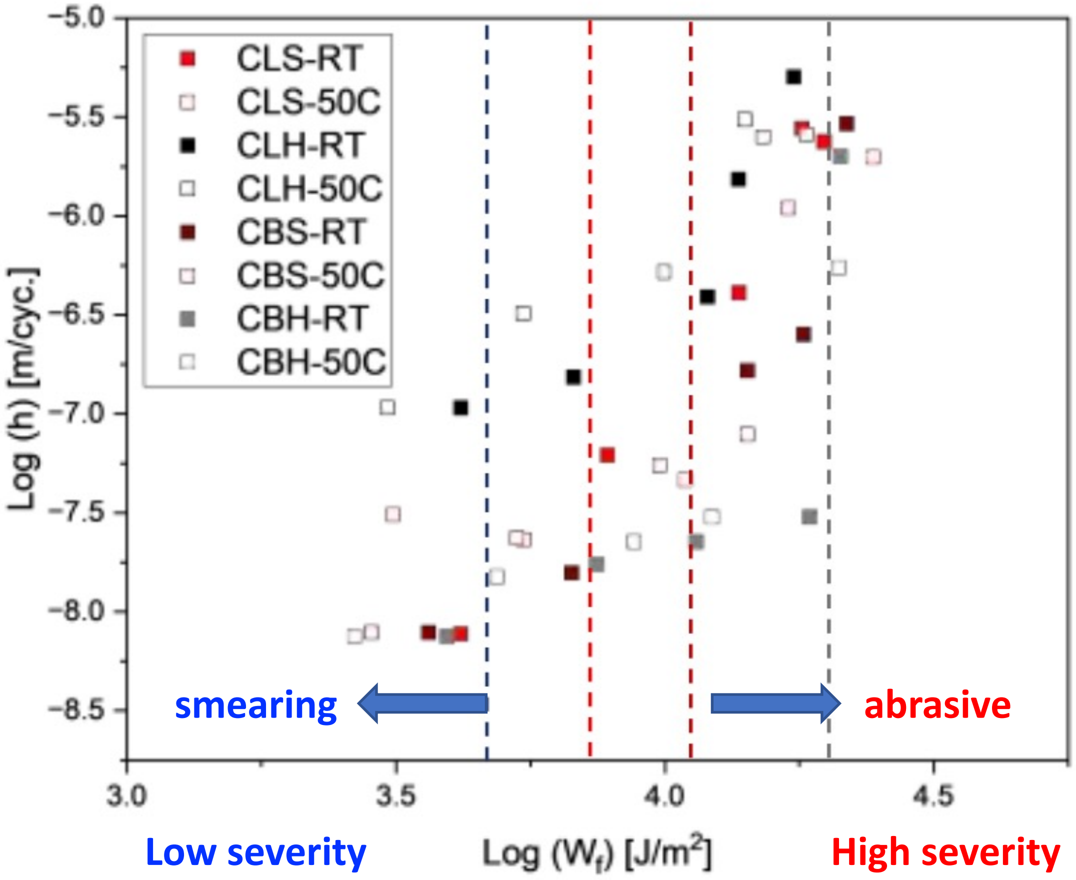
[8] Y. Fukahori, H. Liang, and J. J. C. Busfield, "Criteria for crack initiation during rubber abrasion," *Wear*, vol. 265, no. 3-4, pp. 387-395, 2008, doi: 10.1016/j.wear.2007.11.008.

Conclusion: Transitions for tread compounds / Summary

How does the frictional energy threshold change between the tested tread compounds?

Effect of changing the amount of accelerator (i.e. crosslinking)

- **CLH** suffers from early onset thermo-mechanical degradation due to **weak mechanical properties**
- **CLS** showcases combination of tribo-chemical and thermo-mechanical degradation due **flexible network**



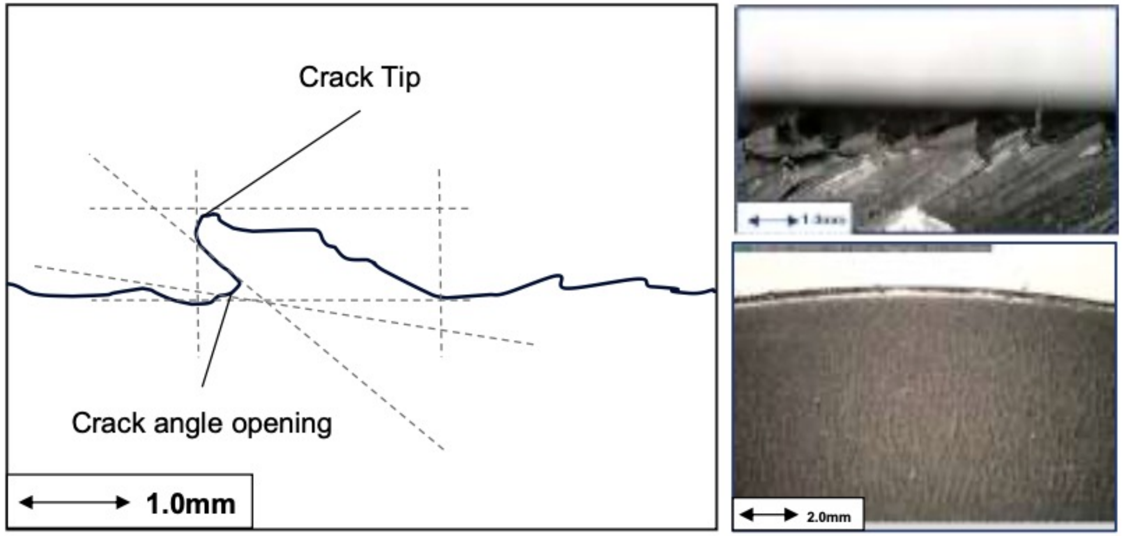
Effect of changing the amount of carbon black

- **CBS** showcases later onset **thermo-mechanical degradation** than CLS due to **higher toughness**
- CBH almost always exhibits tribo-chemical degradation due to highest **energy dissipation** and improved **toughness**

How does the frictional energy threshold influence the transition between wear mechanisms?

Abrasive wear

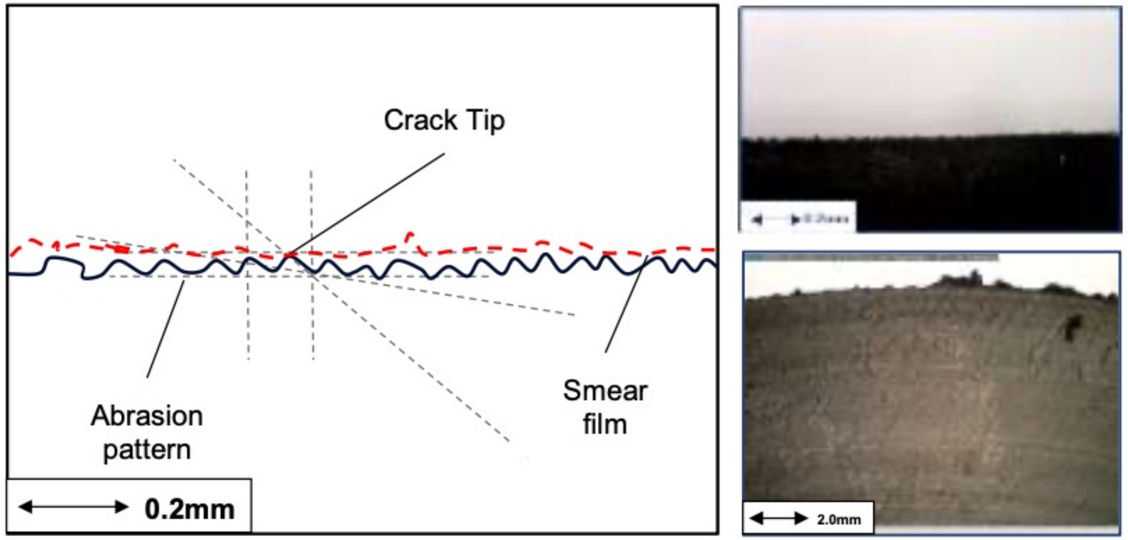
Crack formations develop on the macro-scale at evenly spaced intervals



Thermo-mechanical degradation: Accelerated particle detachment and abrasive (and fatigue) wear

Smear wear

Smaller micro-crack formations occur in much higher frequency



Tribo-chemical degradation: An oxidised sticky layer protects the bulk rubber surface from excessive crack initiation and propagation

Target of Tyre Tread compound Design for reducing TRWP

- **No materials that are harmful to health or the environment**
  Reduce such materials as much as possible

- **Reduce the wear rate to mitigate both volume and size impact of TRWP**
  Design high toughness compound preferring the smearing wear to build the protective layer for wear

- **Keep the better performance of rolling resistance and wet grip of tyre**
  Think the better way to solve the trading off

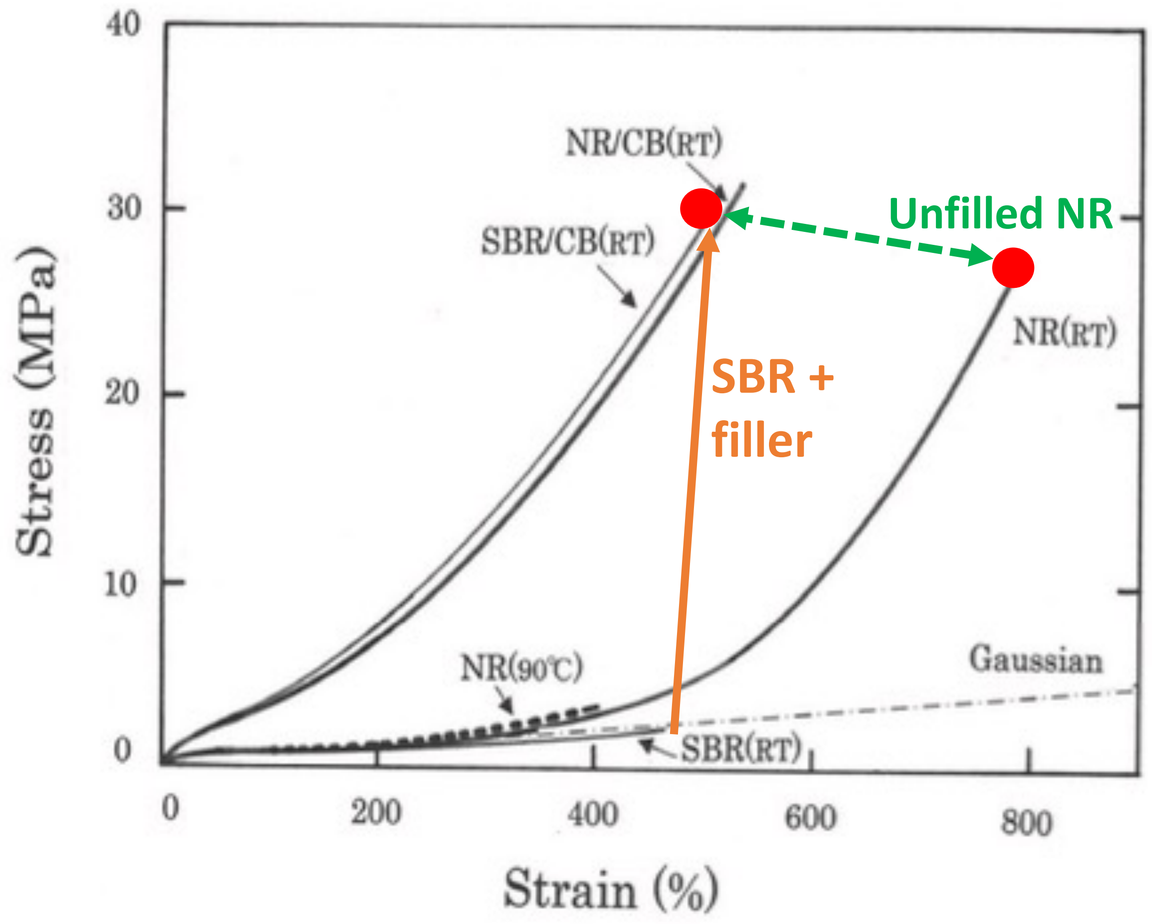
Selection of Polymers

＊NR: natural rubber, IR: Isoprene Rubber, SBR: Styrene Butadiene Rubber, BR: Butadiene Rubber

	NR	IR	SBR	BR
Wear Resistance	◎	○	◎	☆
Wet Traction	○	○	◎	○
Energy Loss (RR)	◎	☆	○	×
Tear Resistance	☆	◎	×	×
Air permeability	×	×	×	×
Heat Degradation Resistance	×	×	○	○

☆Super, ◎ Superior, ○ Fair, × Inferior

Selection of Polymers

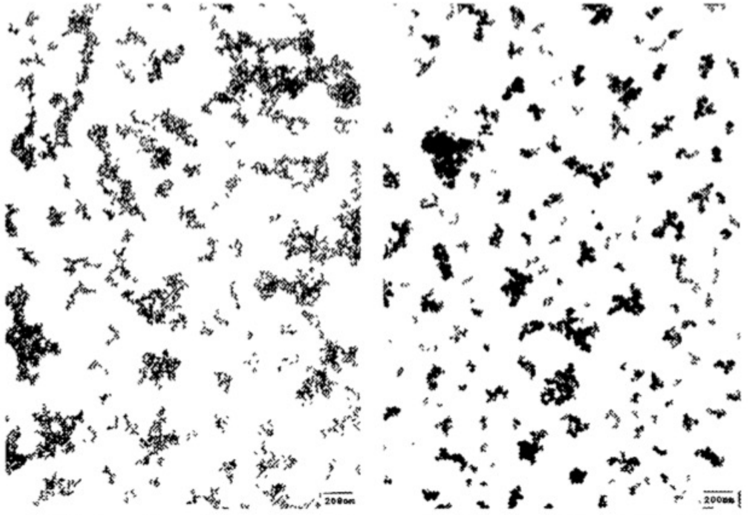


Toughness (Area below stress-strain curve): unfilled NR = filled SBR

Fukahori,Y, Nippon Gomu Kyokaishi, 88, No.8, p.309 (2015)

Selection of Fillers

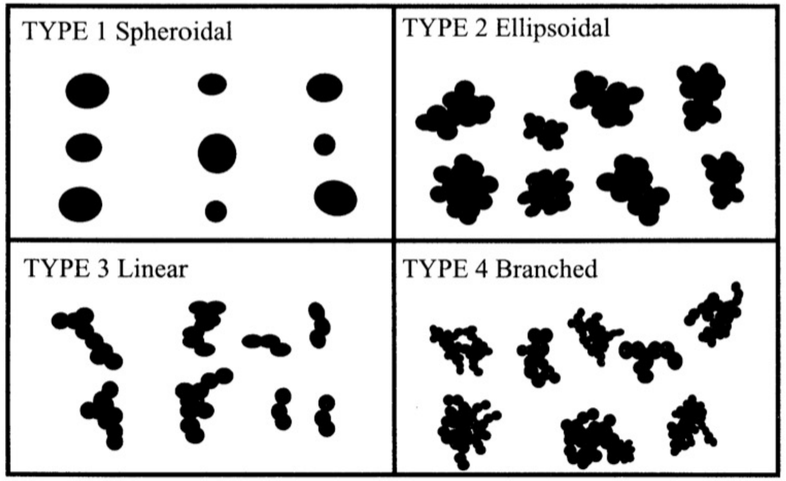
1. Size



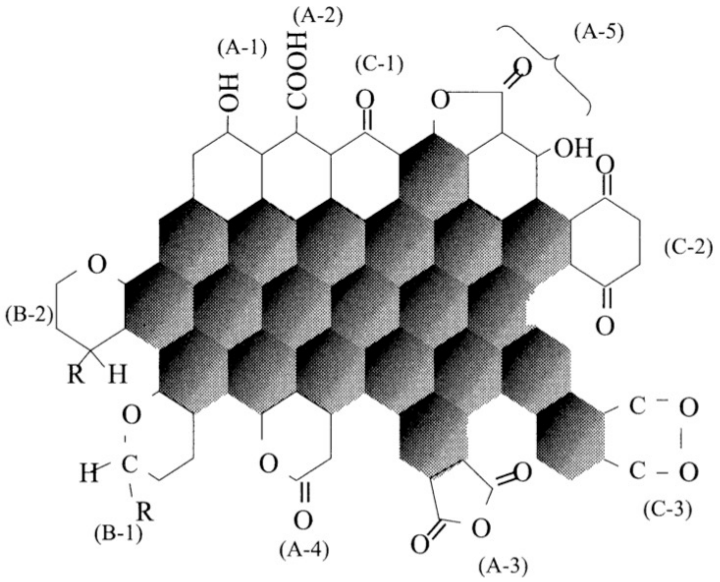
a) High color channel black b) Furnace black (N220)

c) Furnace black (N774) d) Thermal black (N990)

2. Shape



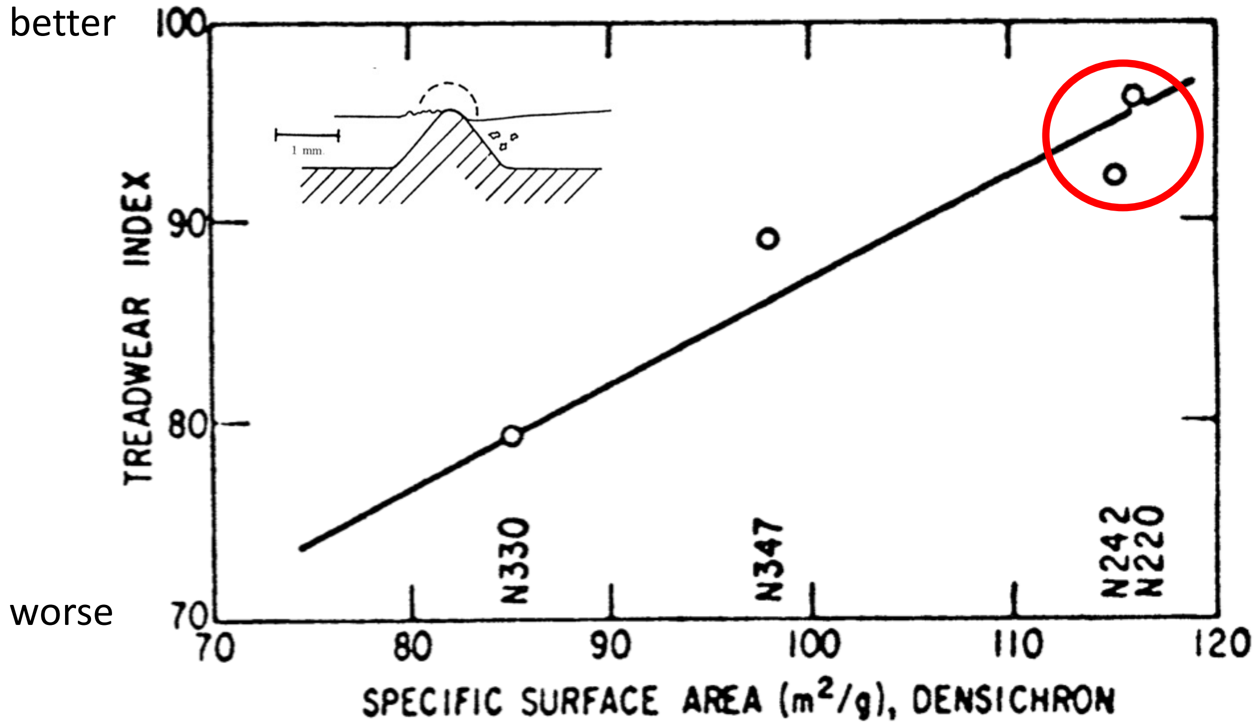
3. Surface

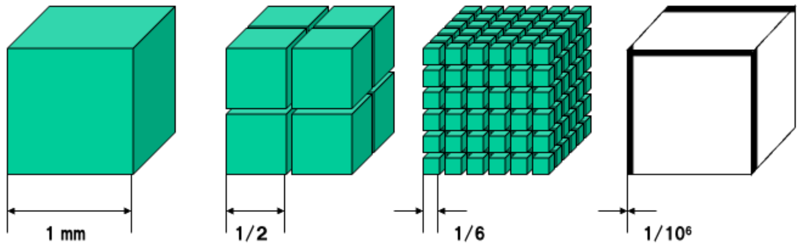


Source: Arai, H, TANSO, N0.223, pp.232-243 (2006)

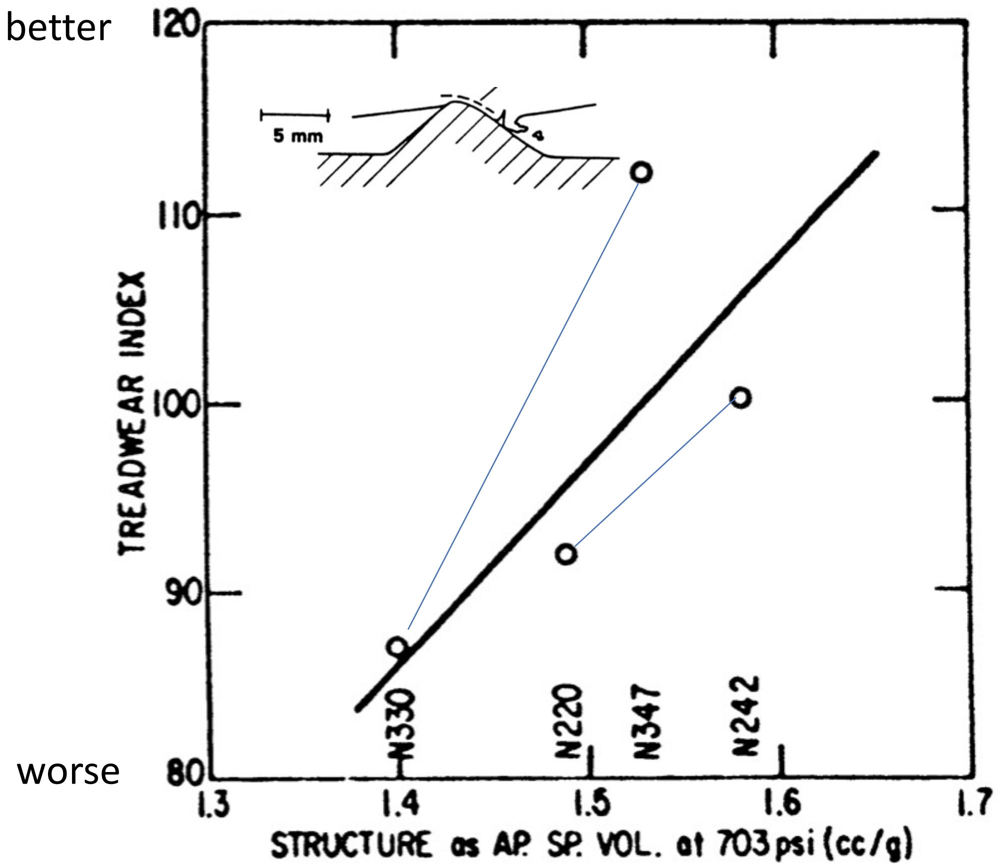
Selection of Fillers

Size effect on Low severity wear



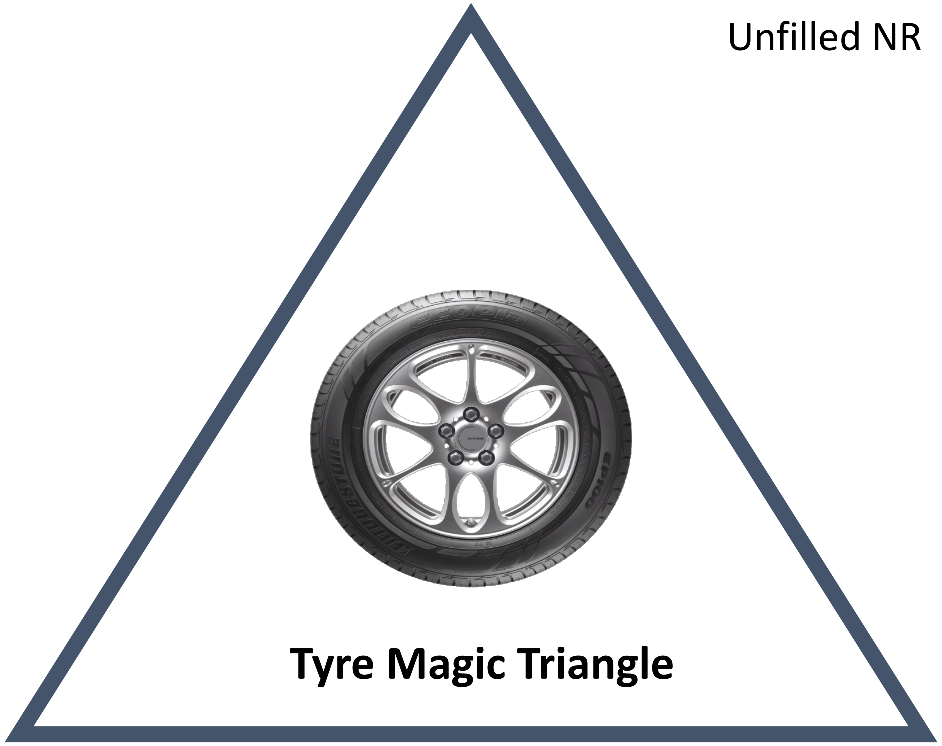


Shape effect on High severity wear



Source: Science and Technology of Rubber(IX) "The Rubber Compound and Its Composition, Rubber Age, 107(8) 20-35, (9)3 9-45(1975)

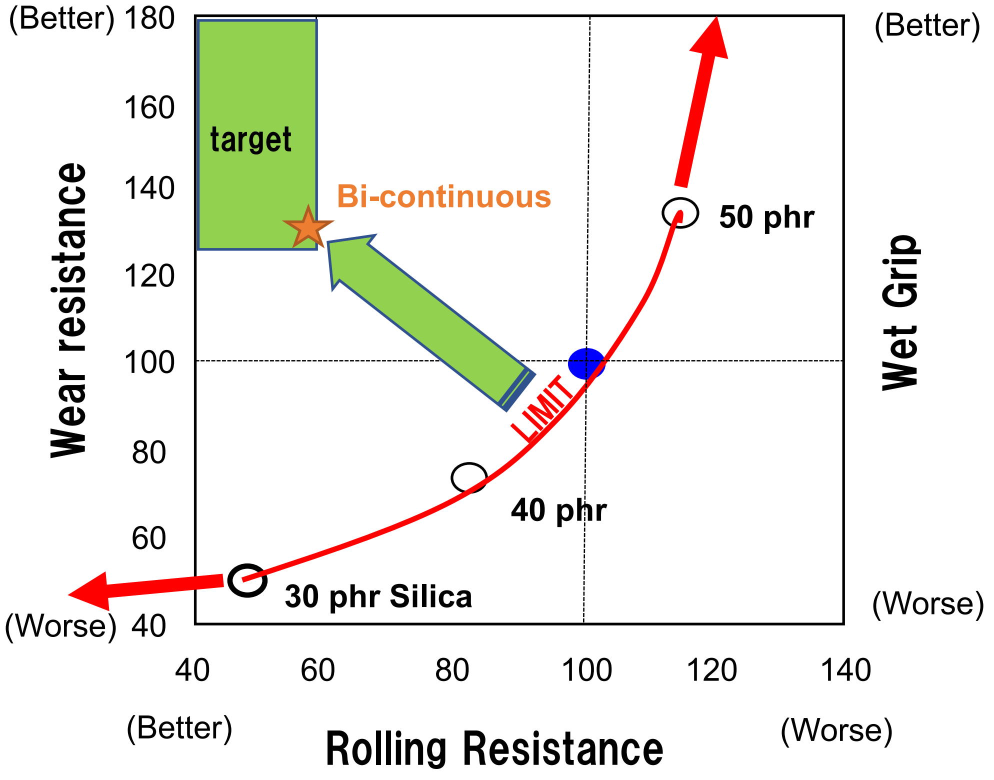
Trading off between Tyre Performances

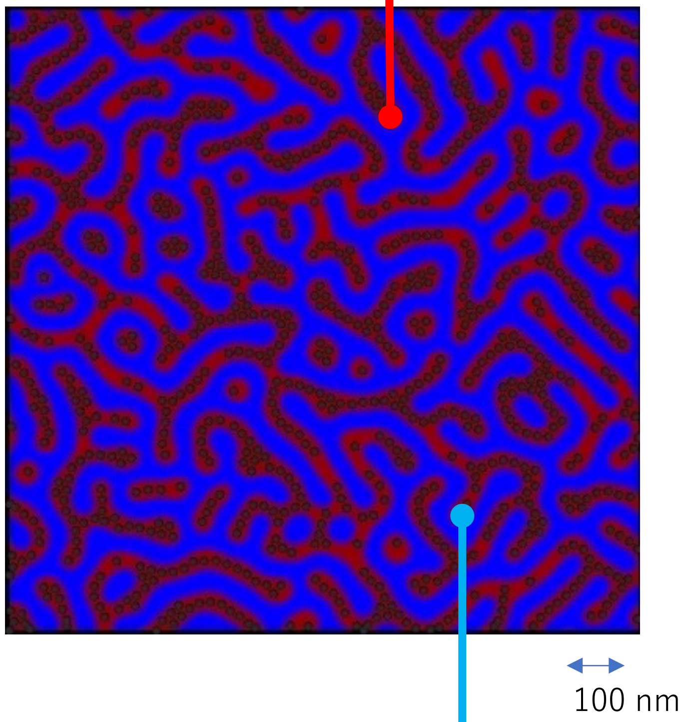


How to design ideal nano structure of blend

High loading filler gives better wear and saturation of energy loss/ WET grip

Ultimate high toughness domain

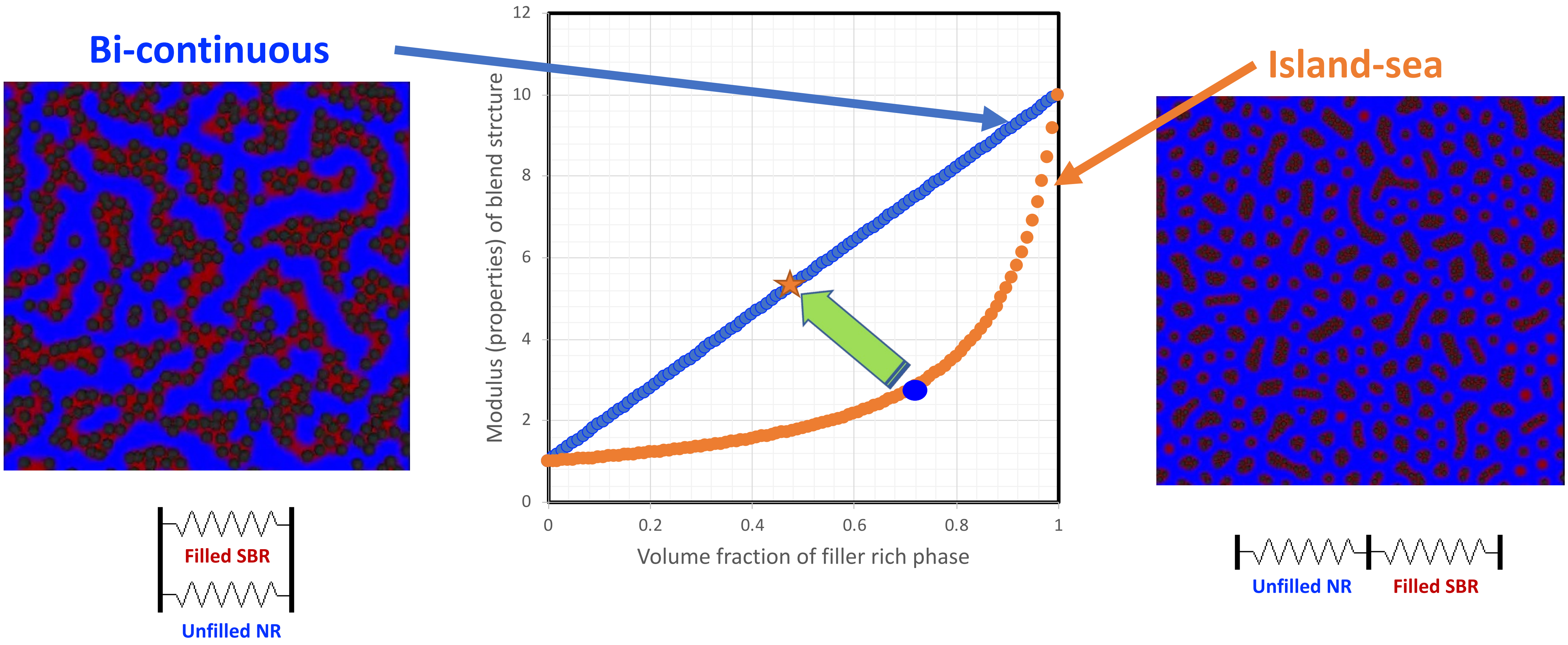




Less loading filler gives better energy loss and saturation of wear

Ultimate low energy loss domain

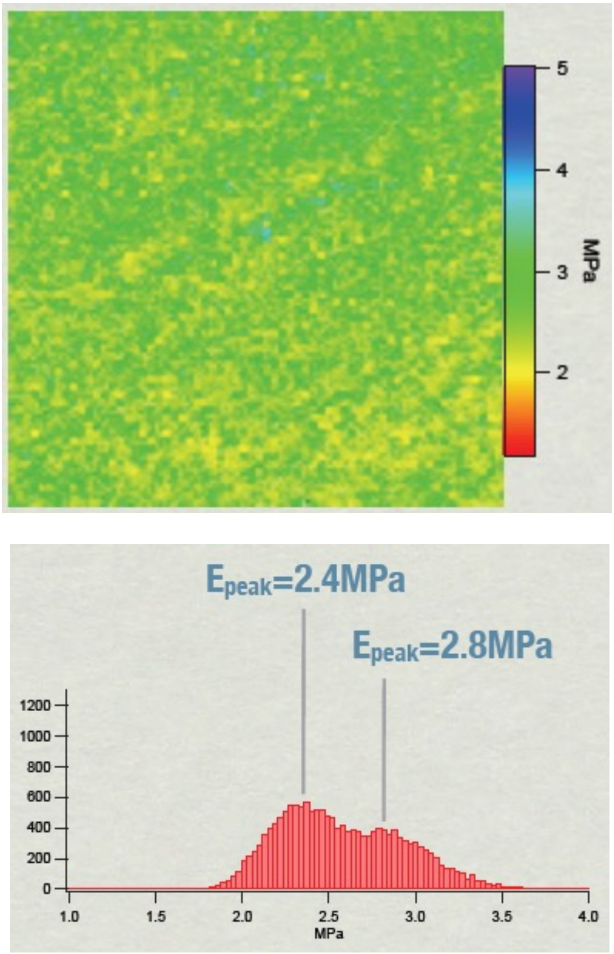
Bi-continuous Polymer Blend Structure



- Highly filled SBR phase can be achieved by the functionalized SBR with Chain End-modification.
- Bi-continuous structure optimizes two extreme trading off performances.

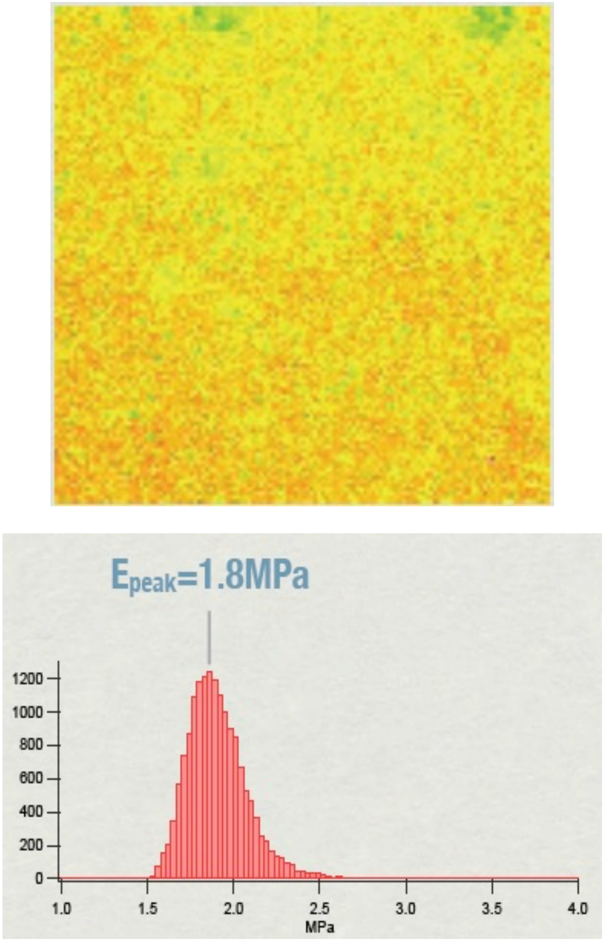
Process for High Toughness and Low Hysteresis Compound: Curing

Conventional cure



AFM modulus mapping

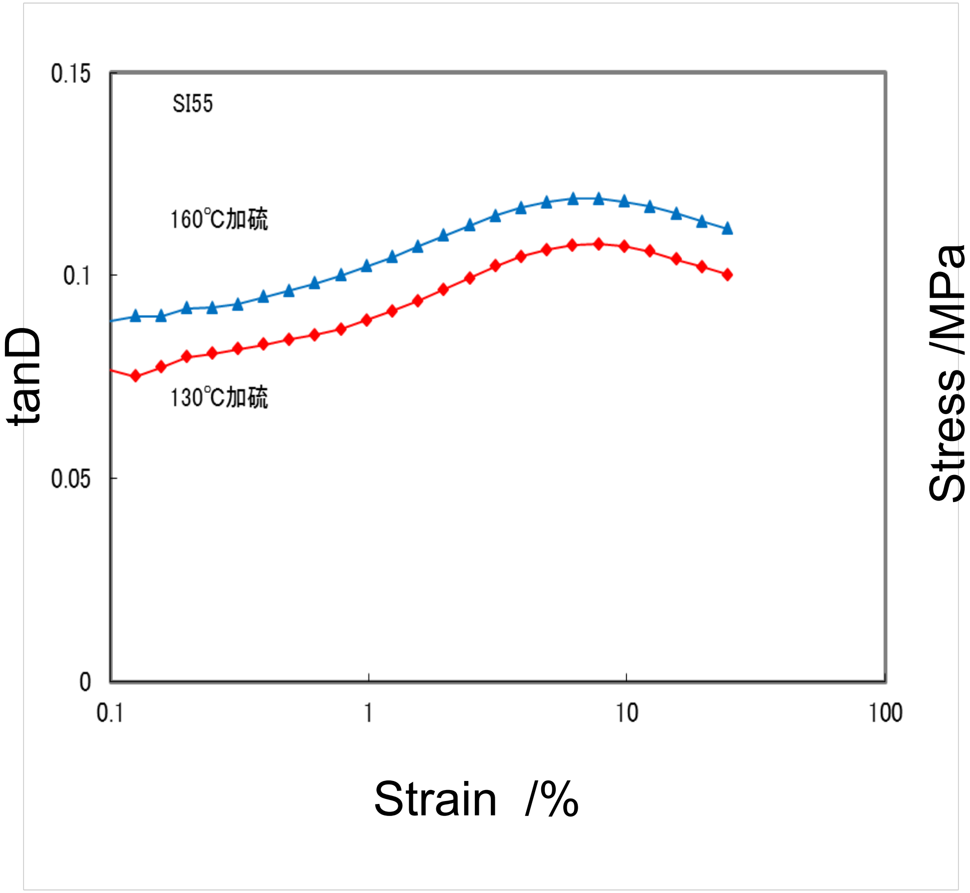
Low temperature cure

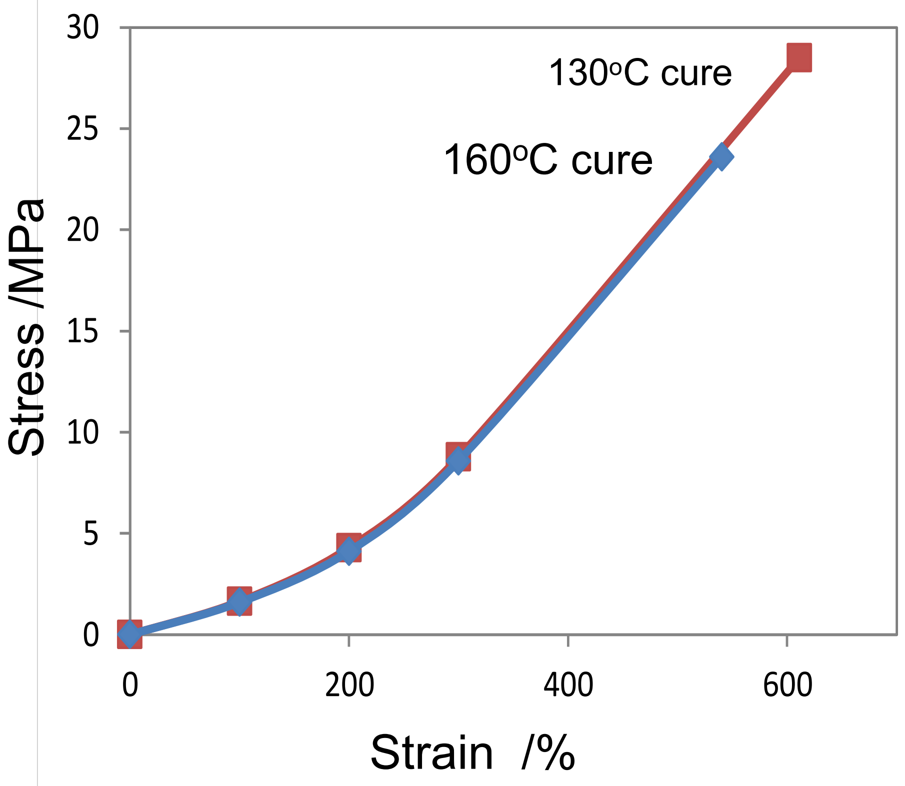


AFM modulus mapping

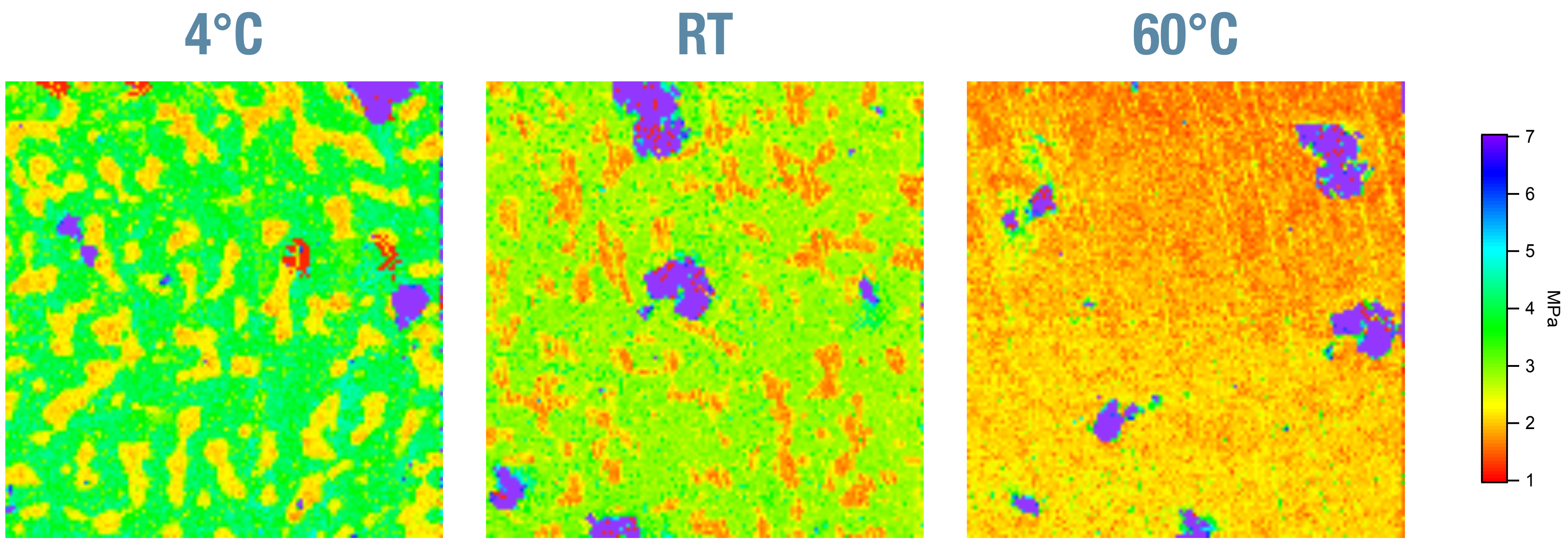
Low temperature cure gives uniform crosslinked network which gives lower tanD and higher toughness

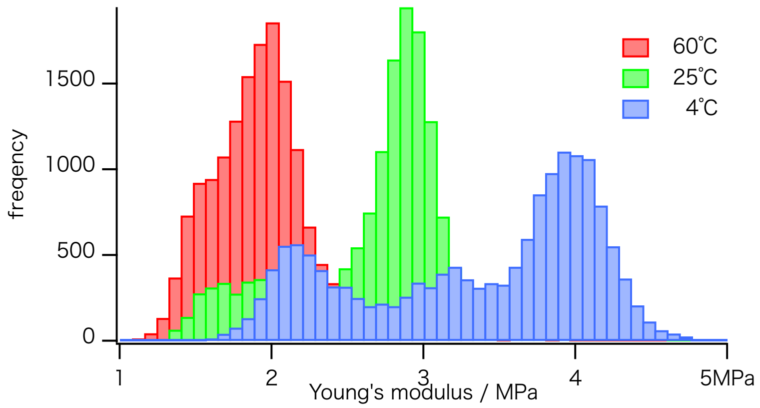
Curing Process for Low Hysteresis and High Toughness

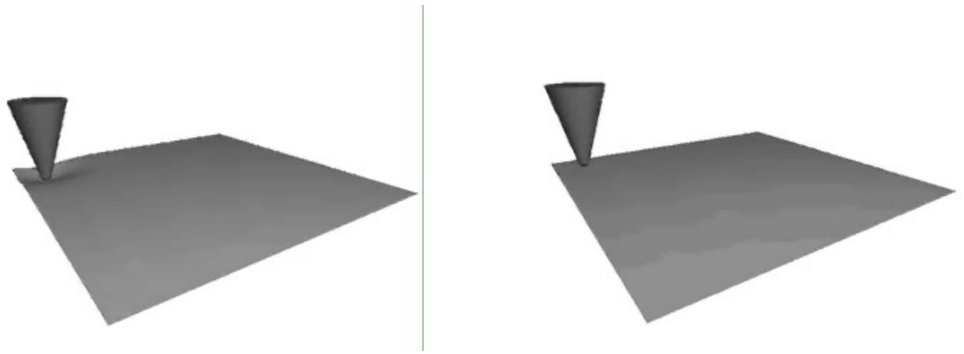




Modulus Mapping on Final Product of Bi-continuous Polymer Blend







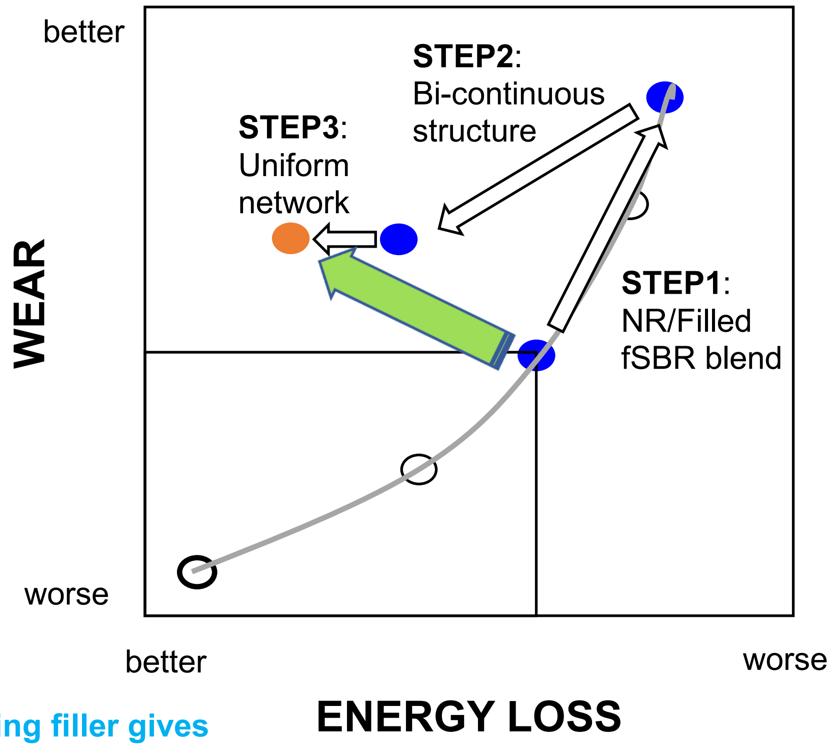
conventional

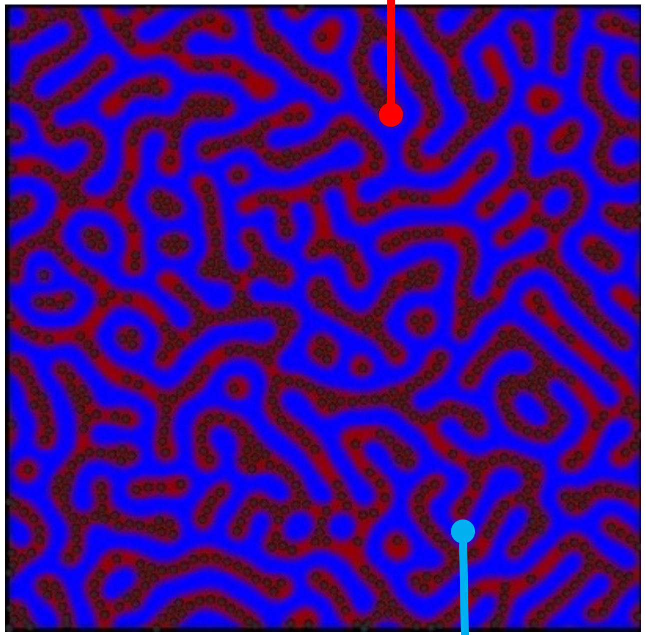
new

Performance Map on Final Product of Bi-continuous Polymer Blend

High loading filler gives better wear and saturation of energy loss/ WET grip

Filled SBR (higher toughness and higher hyseresis)





Less loading filler gives better energy loss and saturation of wear

Unfilled NR (lower energy loss and higher toughness)

Formulation improving Tyre wear performance

materials	Reference	Proposed recipe
ESBR	100	-
End modified SSBR	-	30
NR	-	70
Carbon black (high-surface area & structure)	50	5
Silica (high surface area)		50
Coupling agent		5
Oil	20	TDAE
Anti-oxidant	1	Alternative 6PPD
Zinc oxide	3	Alternative ZnO
Stearic acid	2	←
Accelerator	1.2	←
Sulfur	1.5	←

Summary

Tyre tread compounds that solves TRWP issues keeping the other tyre performances

- Remove the materials which are harmful to human and environment
- Design high toughness compound enhancing smearing to reduce TRWP
- Bi-continuous structure of unfilled NR and highly filled SBR to achieve high toughness